



LMS QUIZ
VII-SEMESTER (CSE Core)
DEEP-LEARNING (CSE_4064)

Duration: 30-minutes

Student name	Reg no.	Section	Semester

Q NO.	Question	Marks
Q1	In the context of learning XOR using a neural network, why is it necessary to use at least one hidden layer? A) XOR cannot be solved using neural networks. B) XOR can be solved using linear classifiers, so no hidden layer is needed. C) XOR is a linear problem, and no hidden layer is required. D) XOR requires a non-linear transformation, which can only be achieved with a hidden layer.	0.5M
Q2	The rank of a matrix represents: A) The determinant of the matrix B) The maximum number of linearly independent row vectors or column vectors C) The size of the matrix D) The number of rows in the matrix	0.5M
Q3	L1 regularization tends to: A) Make the optimization process faster without affecting model accuracy. B) Shrink the weights to zero, performing feature selection. C) Reduce model complexity by increasing bias. D) Increase the magnitude of weights, reducing underfitting.	0.5M
Q4.	Which of the following techniques can help in reducing overfitting in deep learning models? A) Increasing model complexity B) Decreasing training time C) Data augmentation D) Reducing data size	0.5M
Q5	In the context of neural networks, which technique is most commonly used to prevent overfitting when training on limited data? A) Dropout B) Early stopping C) Data augmentation D) All of the above	0.5M
Q6	The LSTM (Long Short-Term Memory) network is particularly well-suited for: A) Tasks that require dense feature representations. B) Problems requiring fixed-length outputs like classification tasks. C) Image recognition tasks. D) Time-series data and sequence prediction problems.	0.5M
Q7	Support Vector Machines (SVMs) with a Radial Basis Function (RBF) kernel are effective in: A) Classification tasks with a simple boundary between classes. B) Predicting continuous values for regression problems. C) Linear regression tasks with high-dimensional data. D) Classifying non-linear data by mapping it into a higher-dimensional space.	0.5M

Q8	In deep learning, why does the Vanishing Gradient Problem occur in very deep networks, and what are the main solutions to mitigate it? A) Due to large weight values in the early layers, leading to exploding gradients; using ReLU activation functions. B) Due to improper initialization of weights; using skip connections and normalization layers. C) Due to noisy data, leading to incorrect gradients; using dropout and early stopping. D) Due to small weight values in the early layers, leading to vanishing gradients; using batch normalization and LSTMs.	0.5M
Q9	In Bayesian statistics, prior probability refers to: A) The initial assumption about the parameters before seeing the data B) The likelihood of the data given the parameters C) The model evaluation D) The posterior distribution	0.5M
Q10	A model that has too few parameters relative to the complexity of the data is: A) Underfitting B) Prone to high variance C) A good model D) Overfitting	0.5M
Q11	XGBoost is a gradient boosting algorithm that: A) Relies on deep neural networks for boosting. B) Utilizes principal component analysis to extract features. C) Uses bagging to create multiple models from subsets of data. D) Minimizes the loss function by building weak learners sequentially.	0.5M
Q12	Maximum Likelihood Estimation (MLE) is used to: A) Calculate the posterior probability B) Minimize the bias of the estimator C) Maximize the entropy of the distribution D) Estimate model parameters that maximize the likelihood of the observed data	0.5M
Q13	In deep learning, what is the key advantage of using pre-trained models? A) Simpler model architectures B) Faster computation and lower accuracy C) Reduction in the need for labelled data D) No need for model fine-tuning	0.5M
Q14	The SoftMax activation function is primarily used in: A) Multi-class classification problems to convert logits to probabilities. B) Binary classification problems with one output unit. C) Sequential data processing tasks like language modelling. D) Regression problems to predict a continuous output	0.5M
Q15	Which of the following methods is used for solving linear systems of equations numerically? A) Support vector machines B) Backpropagation C) Gaussian elimination D) Gradient descent	0.5M

Q16	The success of deep learning in tasks such as image recognition can be attributed to the ability of deep networks to: A) Provide linear separability for complex problems. B) Approximate any continuous function with a small number of layers. C) Learn hierarchical feature representations at multiple levels of abstraction. D) Avoid overfitting by restricting the size of the network.	0.5M
Q17	Which of the following is the correct approach for handling multi-class classification in neural networks? A) Tanh activation function in the output layer B) Softmax activation function in the output layer C) ReLU activation function in the output layer D) Sigmoid activation function in the output layer	0.5M
Q18	L2 regularization (also known as Ridge Regression) works by adding a penalty term to the loss function that is proportional to the: A) Sum of the absolute values of the weights. B) Absolute value of the output. C) Size of the input features. D) Sum of the squares of the weights.	0.5M
Q19	In machine learning, what is the role of a validation set? A) To tune hyperparameters and evaluate performance B) To test the model C) To train the model D) To define the loss function	0.5M
Q20	What is a recent trend in deep learning that addresses its computational expense? A) Batch normalization B) Transfer learning C) Generative models D) Data augmentation	0.5M
Q21	The learning rate schedule used in training deep neural networks typically adjusts the learning rate: A) Linearly, by decreasing over time. B) Constantly, without any changes. C) In a step-wise fashion or decay. D) Exponentially, based on validation loss.	0.5M
Q22	When a model's performance on training data is very high, but its performance on validation data is significantly worse, this is indicative of: A) Overfitting B) Underfitting C) A perfectly tuned model D) Lack of training data	0.5M
Q23	Batch normalization helps to accelerate training by: A) Introducing random noise during training. B) Reducing the dimensionality of the input data. C) Normalizing the data at each layer during training. D) Adjusting the learning rate dynamically.	0.5M

Q24	In deep feedforward networks, what does the backpropagation algorithm compute? A) The gradients of the loss function with respect to the weights of the network. B) The final predicted output of the network. C) The forward pass through the network to compute the output. D) The optimal learning rate for the training process.	0.5M
Q25	Eigenvalues and eigenvectors are essential in deep learning because they: A) Control the scaling of data B) Help with optimization C) Define the geometry of the data D) Improve overfitting issues	0.5M
Q26	A deep feedforward neural network with multiple layers can approximate any continuous function, but which of the following is required for the approximation to be universal? A) The activation function must be non-linear. B) The number of neurons must be small to avoid overfitting. C) The training algorithm must use unsupervised learning. D) The network must have a sufficiently large number of layers and neurons in each layer.	0.5M
Q27	Random Forests work by: A) Using gradient descent to find optimal parameters. B) Using a single decision tree to make predictions. C) Combining decision trees in a random way and using majority voting for classification. D) Reducing overfitting by pruning decision trees during training.	0.5M
Q28	The trade-off between bias and variance is crucial for: A) Data normalization B) Model complexity adjustment C) Data preprocessing D) Model selection and performance evaluation	0.5M
Q29	K-means clustering minimizes the: A) Total variance within each cluster by assigning data points to their nearest centroid. B) Total number of clusters by merging similar groups. C) Distance between all points in the dataset and the centroid. D) Between-cluster variance while maximizing within-cluster variance.	0.5M
Q30	The Kullback-Leibler divergence between two probability distributions P and Q is used to measure: A) The similarity between P and Q B) The difference between P and Q C) The variance of P D) The entropy of P	0.5M